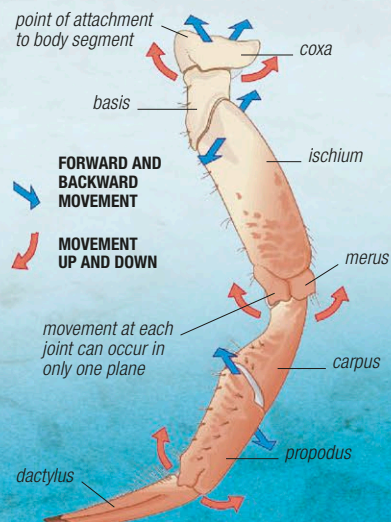


Movement

One of the key features that separates arthropods from most other invertebrates is their jointed limbs. Most arthropods are active animals—moving by walking, swimming, or jumping—which enables them to look for food or mates, colonize new habitats, and escape from predators or adverse local conditions. The number of legs varies from 3 pairs in insects to several hundred pairs in millipedes. Each leg usually ends in a claw of some kind that improves the arthropod's grip on the surface. Many species also have adaptations, such as bristles or adhesive pads, for walking on smooth, vertical surfaces or on water. In some species, the joints between leg segments contain an elastic protein, called resilin. When compressed, this acts as a reserve of energy that can be released when the animal is on the move. In addition to movement, arthropods also use their legs to catch prey, to mate, and to communicate.



LEG MOVEMENT

An arthropod's limb, such as the first walking leg of a crab (left), consists of a set of hollow, rigid sections. These are joined by areas of flexible cuticle and moved by muscles that connect the segments. The ends of the segments are shaped so that movement at each joint occurs in a different plane to movement at the adjacent joints. The result is a limb that can be moved into almost any position.



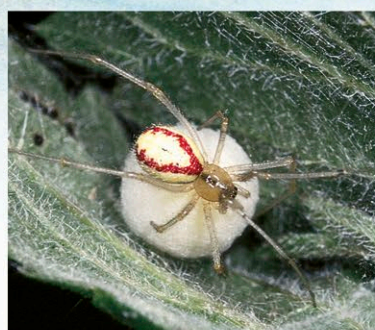
WALKING

Most insects, including this beetle, walk with what is known as a tripod gait. As each leg is lifted off the ground, the body is supported by one leg on the same side of the body and 2 legs on the opposite side.



Growth

One disadvantage of having an exoskeleton is that it must periodically be replaced with a larger one as the animal grows. Arthropods grow and develop in various ways. Crustaceans, myriapods, and arachnids molt throughout their adult life. Larval crustaceans look nothing like the adults. In crabs, for example, the egg hatches into a planktonic zoea larva. This becomes a bottom-dwelling megalopa larva, which in turn develops into the adult. Young myriapods typically look like small adults and become longer by adding trunk segments at each molt, although some centipedes hatch with the adult number of segments. Young arachnids typically look like small versions of their parents. With the exception of bristletails and silverfish, insects stop molting upon reaching sexual maturity.



PARENTAL CARE

It is not unusual for adult insects, centipedes, and arachnids to care for their eggs and young. Female spiders wrap their eggs in silk and carry or guard them until they hatch. Scorpions brood their eggs and carry their young on their back after they hatch.



MOLTING

When molting, an arthropod's body is soft and unprotected. An immature damselfly (known as a nymph) molts several times underwater. When ready for the final molt, it crawls out of the water. Once exposed, it must shed its skeleton and take to the air in a couple of hours if it is to avoid being eaten by a predator.

Feeding

Arthropods eat a wide range of foods. Some eat decaying organic matter (saprophagous), others eat the tissues of other animals and plants. Many species are specialized feeders, the shape of their limbs and mouthparts reflecting the food they eat. Strong claws and jaws can tear and cut tough material, such as skin, cuticle, and plant tissue, while needlelike mouthparts are used to suck up liquids, such as sap, blood, and nectar. Many aquatic crustaceans have featherlike structures on their legs or around the mouth that act like strainers to filter minute organisms or organic particles from water. Most arthropods use saliva to lubricate their food as it passes into the digestive system.



PREDATORS

Most arachnids eat other animals. However, they can ingest only liquid food. Having subdued a cockroach, this scorpion will release digestive enzymes over its body and then consume the resulting liquid.

PLANT EATERS

Herbivorous arthropods can eat large quantities of food. A swarm of locusts can consist of up to 50 billion individuals, theoretically capable of consuming 100,000 tons of food in a day.

